Supplemental Table 3: Characteristics of Included RCTs sorted by treatment time period

Study,Year	Type	Dosage and Timing	Odds Ratio
Burns, 2005	Perioperative	The first 2 doses of N-acetylcysteine or placebo were administered in the operating room immediately following induction of anesthesia and on successful weaning from CPB. The third and fourth doses were administered in the ICU, 12 and 24 hours after administration of the first dose.	1.04 [0.63, 1.72]
Adabag, A. S. 2008	Perioperative	Study medication was started 1 day before surgery and was administered orally twice daily for a total of 14 doses (3 doses before surgery and 11 after the surgery) or until discharge from the hospital (if this occurred before the completion of the 14 doses).	1.36 [0.62,3.02]
Prasad, A. 2010 (done)	Perioperative	The NAC group received oral NAC 600 mg twice a day on the preoperative day and intravenous NAC 600 mg prior to the induction of anaesthesia on the day of surgery. This was followed by intravenous NAC 600 mg twice a day until the second postoperative day (total dose of NAC administered was 4.8 g).	0.73 [0.15,3.51]
Santana 2014	Perioperative	NAC 150 mg/kg in 500 mL in 0.9% IV saline in 2 hours, started 2 hours before surgery, followed by NAC 50 mg/kg in 500 mL 0.9% IV saline over 6 hours (or 0.9% IV saline (control group) q at the same volume	0.30 [0.11,0.81]
Amini 2018	Perioperative	Patients in the selenium, vitamin C, and NAC groups received 0.5 mg, 1500 mg, and 600 mg tablets, respectively, twice a day, from 24 hours before the operation until two postoperative days.	1.73 [0.72,4.17]
Ristikankare , A. 2006	Intraoperative to Postoperative	NAC was administered in saline 0.9% as a loading dose of 150 mg/kg in 15 min, followed by 50 mg/kg for the next 4 h, and thereafter, 100 mg/kg for 16 h.	0.77 [0.31, 1.88]
Haase, M 2007	Intraoperative to Postoperative	Immediately after the induction of anesthesia, before the first surgical incision, NAC was administered in an intravenous dose of 150 mg/kg in 200 mL of 5% glucose over 15 mins, followed by continuous intravenous infusion of 50 mg/kg in 500 mL of 5% glucose over 4 hrs, then 100 mg/kg in 1 L of 5%	0.87 [0.31,1.88]

		glucose over 20 hrs (total dose, 300 mg/kg body weight over 24 hrs)	
Sisillo, E. 2008	Intraoperative to Postoperative	Patients of the NAC group received an intravenous bolus of 1200 mg of NAC immediately before induction of anesthesia, followed by three additional boluses administered at 12-hr intervals in the ICU (total dose of NAC = 4800 mg)	0.62 [0.38,1.02]
Song, J. W. 2015	Intraoperative to postoperative	Immediately after anaesthetic induction, patients in the NAC group received 150 mg/kg of NAC IV bolus over 15 min, followed by a continuous infusion at 150 mg/kg/day for 24 h.	0.86 [0.40,1.85]
Aldemir, M. 2016	Intraoperative to Postoperative	NAC was administered in 0.9% saline as a loading dose of 150 mg/kg over 15 min, followed by 50 mg/kg over the next 4 h and 100 mg/kg over 16h. The placebo group received similar volumes of 0.9% saline (1).	0.3 [0.21,1.88]